Q1

```
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test
My Turn from Child Thread
My Turn from Child Thread
Your Turn from Parent Thread
My Turn from Child Thread
Your Turn from Parent Thread
My Turn from Child Thread
My Turn from Child Thread
Your Turn from Parent Thread
My Turn from Child Thread
My Turn from Child Thread
Your Turn from Parent Thread
My Turn from Child Thread
My Turn from Child Thread
Your Turn from Parent Thread
Parent: Child thread has been successfully canceled andjoined.
(.venv) asjadraza@Asjads-MacBook-Air-2 OS %
```

Q2

```
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test
My Turn from Child Thread 1
My Turn from Child Thread 2
My Turn from Child Thread 3
Your Turn from Parent Thread 1
My Turn from Child Thread 4
My Turn from Child Thread 5
My Turn from Child Thread 6
My Turn from Child Thread 7
Your Turn from Parent Thread 2
My Turn from Child Thread 8
My Turn from Child Thread 9
My Turn from Child Thread 10
Your Turn from Parent Thread 3
Parent: Child thread has been successfully canceled andjoined.
(.venv) asjadraza@Asjads-MacBook-Air-2 OS %
```

Q3

Without `pthread_join(newthread, NULL);`

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test
My Turn from Child Thread 1
Your Turn from Parent Thread 1
My Turn from Child Thread 2
My Turn from Child Thread 3
Your Turn from Parent Thread 2
My Turn from Child Thread 4
My Turn from Child Thread 5
Your Turn from Parent Thread 3
Parent: Child thread has been successfully canceled and joined.
asjadraza@Asjads-MacBook-Air-2 OS %
```

This is because without pthread_join the main thread completes its execution and exits without waiting for the child thread, which is why child could only complete its 11 iterations. However with pthread_join parent waits till child thread finishes.

Placing `pthread_join(newthread, NULL); right after yourturn();`

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test
My Turn from Child Thread 1
My Turn from Child Thread 2
Your Turn from Parent Thread 1
My Turn from Child Thread 3
My Turn from Child Thread 4
Your Turn from Parent Thread 2
My Turn from Child Thread 5
Your Turn from Parent Thread 3
My Turn from Child Thread 6
My Turn from Child Thread 7
My Turn from Child Thread 8
My Turn from Child Thread 9
My Turn from Child Thread 10
Parent: Child thread has been successfully canceled and joined.
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q4

```
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test
Thread 0: Thread ID: 6136524800
Thread 4: Thread ID: 6138818560
Thread 1: Thread ID: 6137098240
Thread 2: Thread ID: 6137671680
Thread 3: Thread ID: 6138245120
Main: All threads are done. Program exiting.
(.venv) asjadraza@Asjads-MacBook-Air-2 OS %
```

Q5

```
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test 4
Given number: 4
The square of the number is: 16
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test 1231908
Given number: 1231908
The square of the number is: 1473864976
(.venv) asjadraza@Asjads-MacBook-Air-2 OS % gcc test.c -o test -lpthread && ./test -123
Given number: -123
The square of the number is: 15129
(.venv) asjadraza@Asjads-MacBook-Air-2 OS %
```